Agentic AI for Automated Gas-Flow Simulation & Optimisation

Independent Python project | September 2026 | Research-and-engineering portfolio proof of concept

Project question

Can a goal-driven, machine-learning-guided agent automatically propose chamber-flow configurations, validate inputs, call a numerical physics tool, learn from observed results and report improved flow uniformity with traceable evidence?

Agent workflow

PLAN	VALIDATE	SIMULATE	LEARN	REPORT
Admissible design	Geometry constraints	Finite-volume model	RF surrogate	Metrics & plots

Reference results (seed 42, grid 60 x 36, 23 evaluated configurations)

Metric	Chosen baseline	Selected design
Flow non-uniformity (CV)	0.3384	0.0741
Normalised outlet flux	0.4446	0.6490
Objective (smaller is better)	0.5912	0.0753

Measured relative CV decrease: 78.1% against this intentionally unoptimised starting geometry. This is a reduced-order benchmark, not a result measured on real semiconductor equipment.

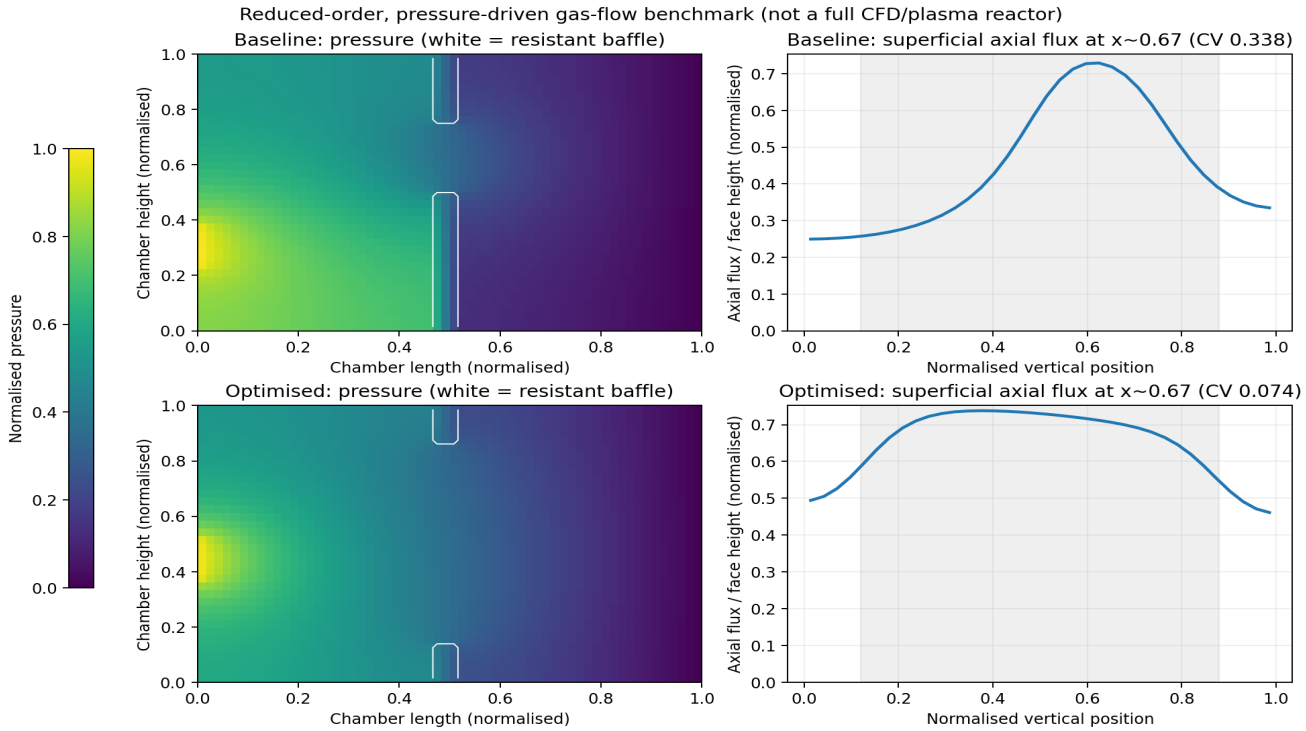


Figure: comparison of normalised pressure fields and axial flux distributions. A low-permeability baffle is outlined in white.

Research motivation and technical method

Why I started this project

With a Master's degree in Computer Science focused on AI and Machine Learning, I wanted to apply AI to an industrial problem with measurable engineering outcomes. The University of Exeter and Oxford Instruments Plasma Technology KTP challenge highlighted the repetitive setup, execution and interpretation involved in CFD workflows. I initiated this independent prototype to test whether an autonomous, auditable ML-guided agent could help coordinate those steps, while keeping safety bounds and numerical checks explicit.

What I actually implemented

Physics tool: a conservative 2D cell-centred finite-volume solver for a steady pressure-driven Darcy/lubrication-type approximation; variable inlet opening and resistive baffle; mass-balance checks.

Agent and ML: feasibility checking, bounded exploration, Random Forest surrogate trained from solver feedback, uncertainty-aware candidate selection, persistent experiment log and report generation. It uses no external LLM or proprietary industrial data.

Evaluation: flux coefficient of variation over a specified sampling band and a penalty for insufficient total flow. The best candidate is chosen using actual physics-tool evaluations, not surrogate predictions alone.

Why the model is deliberately simplified

A lightweight solver lets me demonstrate, test and explain the closed-loop AI architecture before connecting it to a complex external CFD environment. The current prototype is not a Navier-Stokes simulation, a full plasma reactor model, experimental validation or evidence of an industrial productivity gain.

Next research and engineering steps

Integrate an approved OpenFOAM/in-house CFD template behind a restricted solver interface; agree objective metrics and safe design constraints with engineers; establish mesh and solver-convergence studies; compare simulated flow and process outcomes with trusted reference data; introduce human sign-off for deployment.

Quick portfolio demonstration

```
python -m pip install -r requirements.txt
python run_demo.py --iterations 22 --seed 42
python -m pytest -q
```

The accompanying ZIP includes runnable source, documentation, simulation outputs, trace logs, figures, tests, a motivation statement and interview preparation.

One-sentence interview explanation

I built an auditable AI/ML-driven experiment agent that repeatedly proposes feasible chamber configurations, runs a simplified numerical gas-flow model, validates its outcomes, and learns which simulation to try next - a first step toward automation of verified industrial CFD workflows.